Complex Analysis

Contents

1 Complex Number System

Definition 1.1 **Complex Numbers**

The set of complex numbers $\mathbb{C}$ is $\mathbb{R}^2$, together with the operations

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$$

and

$$(x_1, y_1) \cdot (x_2, y_2) = (x_1 x_2 - y_1 y_2, x_1 y_2 + x_2 y_1)$$

Theorem 1.1 **Field Structure**

$\mathbb{C}$ is a field. With $0 = (0, 0),\ -(x, y) = (-x, -y),\ 1 = (1, 0)$ and

$$(x, y)^{-1} = \left(\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2}\right) \text{ for } (x, y) \neq (0, 0)$$

Notation

If $z = (x, y)$ we call x the "real part" of z and write $x = \mathrm{Re}(z)$.
We call y the "imaginary part" of z and write $y = \mathrm{Im}(z)$.
Define $i = (0, 1)$.

Remark 1.1

There is an embedding

$$\begin{aligned} \mathbb{R} &\hookrightarrow \mathbb{C} \\ x &\longmapsto (x, 0) \end{aligned}$$

so we view $\mathbb{R} \subset \mathbb{C}$.

Notation

Under the above embedding, any $z \in \mathbb{C}$ can be written as $z = \mathrm{Re}(z) + \mathrm{Im}(z) \cdot i$
and $i^2 = -1$

Definition 1.2 **Conjugation**

Define *conjugation* to be the map

$$\begin{aligned} \bar{\ } : \mathbb{C} &\longrightarrow \mathbb{C} \\ z = x + yi &\longmapsto x - yi = \overline{z} \end{aligned}$$

Note

$$\mathrm{Re}(z) = \frac{z + \overline{z}}{2}, \quad \mathrm{Im}(z) = \frac{z - \overline{z}}{2i}$$

Note

If $z = x + yi \in \mathbb{C}$ it has a polar representation $(x, y) \mapsto (r, \theta)$ via $z = r\cos\theta + ir\sin\theta$ where $r = \sqrt{x^2 + y^2} \geq 0, x = r\cos\theta, y = r\sin\theta$. The angle θ is defined modulo 2π when $z \neq 0$, and is arbitrary when $z = 0$.

Remark 1.2

If z_1, z_2 have polar representations

$$z_1 : (r_1, \theta_1), \quad z_2 : (r_2, \theta_2)$$

Then

$$\begin{aligned} z_1 \cdot z_2 &= (r_1 \cos\theta_1 + ir_1 \sin\theta_1) \cdot (r_2 \cos\theta_2 + ir_2 \sin\theta_2) \\ &= r_1 r_2((\cos\theta_1 \cos\theta_2 - \sin\theta_1 \sin\theta_2) + i(\cos\theta_1 \sin\theta_2 + \sin\theta_1 \cos\theta_2)) \\ &= r_1 r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)) \end{aligned}$$

Note

If $z = x + iy$ and $r = \sqrt{x^2 + y^2}$, then $r^2 = z\overline{z}$.

Remark 1.3

$\mathbb{C}$ is a metric space and thus the triangle inequality holds.

Remark 1.4

$\mathbb{C}$ is a topological space, so topological notions make sense.

Definition 1.3 **Argument**

If $z \in \mathbb{C} \setminus \{0\}$ has polar representation (r, θ), we call θ the *argument* of z and write $\theta = \arg(z)$. We'll think about arg as a multivalued function. We write $\mathrm{Arg}(z) = \theta$ if $\theta \in (-\pi, \pi]$, the *principal value*.

Example 1.1

If $z = r(\cos\theta + i\sin\theta)$ and $z \neq 0$ then we can find an n^{th} root of z (actually n of them).

2 Differentiation

Definition 2.1 **Complex Derivative**

Suppose $f : U \to \mathbb{C}$ where $U \subseteq \mathbb{C}$ is open and contains z_0. We'll say that f has derivative $f'(z_0)$ at z_0 if

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

Theorem 2.1 **Product Rule**

If f and g are differentiable at z_0 then so is fg and

$$(f \cdot g)'(z_0) = f'(z_0)g(z_0) + f(z_0)g'(z_0)$$

Remark 2.1

If $f : \mathbb{C} \to \mathbb{C}$, then we may write $f(z) = u(z) + iv(z)$ where $u, v : \mathbb{C} \to \mathbb{R}$. If f is differentiable at $z_0 = x_0 + iy_0$ (say $f'(z_0) = a + ib$), then

$$\begin{aligned} R(z) &= f(z) - f(z_0) - f'(z_0)(z - z_0) \\ &= u(z) - u(z_0) - a(x - x_0) + b(y - y_0) + i(v(z) - v(z_0) - b(x - x_0) - a(y - y_0)) \\ &= R_1(z) + iR_2(z) \end{aligned}$$

Then $\lim_{z \to z_0} R_1 \frac{z}{|z - z_0|} = \lim_{z \to z_0} R_2 \frac{z}{|z - z_0|} = 0$. Thus both u and v are differentiable and $\frac{\partial u}{\partial x}(z_0) = a$, $\frac{\partial u}{\partial y}(z_0) = -b$, $\frac{\partial v}{\partial x}(z_0) = b$, $\frac{\partial v}{\partial y}(z_0) = a$.

Theorem 2.2 **Cauchy-Riemann Equations**

If $f : \mathbb{C} \to \mathbb{C}$, $f(z) = u(z) + iv(z)$, then f is differentiable at z_0 (as a complex function) iff both u and v are differentiable at z_0 and

$$\begin{gathered} \frac{\partial u}{\partial x}(z_0) = \frac{\partial v}{\partial y}(z_0) \quad \text{and} \\ \frac{\partial v}{\partial x}(z_0) = -\frac{\partial u}{\partial y}(z_0) \end{gathered}$$

These are called the Cauchy-Riemann equations.

Note

If $u : \mathbb{R}^2 \to \mathbb{R}$, then u is differentiable at (x_0, y_0) if $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$ exist and are continuous in neighborhoods of (x_0, y_0).

Definition 2.2 **Exponential function**

The exponential function is defined to be the function

$$\exp(x+iy) = e^x(\cos y + i\sin y)$$

we write $e^z := \exp(z)$.

Remark 2.2

For $y \in \mathbb{R}$, we have $e^{iy} = \cos(y) + i\sin(y)$.

Definition 2.3 **Holomorphic**

If $f : U \to \mathbb{C}$ with $U \subseteq \mathbb{C}$ open, such that f is differentiable at each point in U, then f is *holomorphic*.

Definition 2.4 **Entire**

$f : \mathbb{C} \to \mathbb{C}$ is called *entire* if it is holomorphic on all of $\mathbb{C}$.

Definition 2.5 **Logarithm**

A complex number w is a *logarithm* of z if $e^w = z$.

Remark 2.3

The range of the exponential is $\mathbb{C} \setminus \{0\}$. So there is no logarithm of 0.

Definition 2.6 **Branch of Logarithm**

If $G \subseteq \mathbb{C} \setminus \{0\}$ is a non-empty open set, then a *branch of* log is a continuous function $\ell : G \to \mathbb{C}$ such that for all $z \in G$, $\ell(z)$ is a logarithm of z, i.e. $e^{\ell(z)} = z$

Note

A branch of log may or may not exist depending on the shape of G.

Example 2.1

$G := \mathbb{C} \setminus (-\infty, 0]$. $\ell(z) = \ln|z| + i\operatorname{Arg}(z)$. Then

$$\begin{aligned} e^{\ell(z)} &= e^{\ln|z|}(\cos(\operatorname{Arg}(z)) + i\sin(\operatorname{Arg}(z))) \\ &= |z|(\cos(\operatorname{Arg}(z)) + i\sin(\operatorname{Arg}(z))) \\ &= z \end{aligned}$$

This is called the *principal branch of* log and is denoted by $\operatorname{Log}(z)$

Note

If we define $\ell(z) = \text{Log}(z) + i2\pi k$ on $\mathbb{C} \setminus (-\infty, 0]$ for fixed $k \in \mathbb{Z}$, then we get a branch of log.

Remark 2.4

It is not possible to find a branch of log on $\mathbb{C} \setminus \{0\}$.

Exercise 2.5

In polar form, the Cauchy-Riemann equations become, if $f(z) = u(z) + iv(z)$

$$\begin{gathered} ru_r = v_\theta \\ \text{and } u_\theta = -rv_r \end{gathered}$$

Theorem 2.3 **Derivative of Logarithm**

If $\ell : G \to \mathbb{C}$ is a branch of log, then ℓ is holomorphic and $\ell'(z) = \frac{1}{z}$.

Theorem 2.4 **Polynomial Holomorphicity Criterion**

Suppose f is a polynomial in variables x and y (with complex coefficients), then f is holomorphic iff f can be written as polynomial in $z = x + iy$.

3 Series

Definition 3.1 **Series**

A *series* is a formal sum $\sum_{n=0}^{\infty} c_n$ where the terms $c_n \in \mathbb{C}$. We say a series *converges* if the sequence of partial sums converge otherwise we say it *diverges*.

Lemma 3.1 **Term Test**

If $\sum_{n=0}^{\infty} c_n$ converges, then $c_n \to 0$.

Definition 3.2 **Absolute Convergence**

A series $\sum_{n=0}^{\infty} c_n$ *converges absolutely* if $\sum_{n=0}^{\infty} |c_n| < \infty$

Lemma 3.2 **Absolute Convergence Test**

If $\sum_{n=0}^{\infty} c_n$ converges absolutely, then $\sum_{n=0}^{\infty} c_n$ converges and $\left|\sum_{n=0}^{\infty} c_n\right| \leq \sum_{n=0}^{\infty} |c_n|$

Example 3.1 **Geometric Series**

$\sum_{n=0}^{\infty} c^n$ converges iff $|c| < 1$. If $|c| < 1$ then $\sum_{n=0}^{\infty} c^n = \frac{1}{1-c}$

Definition 3.3 **Convergence of Functions**

Suppose $(f_n)_{n=1}^{\infty}$ is a sequence of $\mathbb{C}$-valued functions on $G \subseteq \mathbb{C}$.

- $(f_n)_{n=1}^{\infty}$ *converges pointwise* to f on $S \subseteq G$ if $\lim_{n\to\infty} f_n(z) = f(z)$ for every $z \in S$.
- $(f_n)_{n=1}^{\infty}$ *converges uniformly* to f on S if $\sup_{z\in S}|f_n(z) - f(z)| \to 0$
- $(f_n)_{n=1}^{\infty}$ *converges locally uniformly* to f on G if for each $z \in G$, there is some open ball $B \subseteq G$ with centre z such that $(f_n)_{n=1}^{\infty}$ converges uniformly to f on B.

Definition 3.4 **Series of Functions**

If $(f_n)_{n=0}^{\infty}$ is a sequence of functions, $\sum_{n=0}^{\infty} f_n$ is the associated series of functions. The series *converges* to f if the partial sums do.

Example 3.2

$z_0 \in \mathbb{C}$ is fixed. $(f_n)_{n=0}^{\infty}$, where $f_n(z) = a_n(z - z_0)^n$ we get the series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$. The power series centred at z_0.

Proposition 3.3 **Region of Convergence**

The region of convergence for a power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ is of the form either

1. $\{z_0\}$
2. $\mathbb{C}$
3. The points in some disk centred at z_0 and possibly points on the boundary.

Example 3.3

$\sum_{n=0}^{\infty} z^n$ has radius of convergence 1. This represents the function $\frac{1}{1-z}$ on $\mathbb{D}$. This converges locally uniformly.

Theorem 3.4 **Comparison Test for Power Series**

Suppose $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ and $\sum_{n=0}^{\infty} b_n(z - z_0)^n$ are power series with radius of convergence R_1, R_2 respectively. If $|b_n| \leq M|a_n|$ for some $M > 0$, then $R_1 \leq R_2$

Proof: This follows from the previous proposition because if $|z - z_0| < R$, where R is the radius of convergence, then $\sum a_n(z - z_0)^n$ converges absolutely. $\square$

Theorem 3.5 **Cauchy-Hadamard Theorem**

The radius of convergence for a power series $\sum a_n(z - z_0)^n$ is

$$\frac{1}{\limsup_{n\to\infty} |a_n|^{\frac{1}{n}}}$$

Theorem 3.6 **Differentiation of Power Series**

Suppose $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ has radius of convergence $R > 0$. Let f be the function it represents (on that disk). Then the radius of convergence for $\sum_{n=1}^{\infty} n a_n (z - z_0)^{n-1}$ is also R and if g is the function it represents, then f is differentiable and $f'(z) = g(z)$ for
$z \in G = \{z \in \mathbb{C} \mid |z - z_0| < R\}$

Proof: Note

$$\begin{aligned}
&\frac{1}{\limsup_{n\to\infty} |n a_n|^{\frac{1}{n}}} \\
&= \frac{1}{\lim n^{\frac{1}{n}}} \cdot \frac{1}{\limsup |a_n|^{\frac{1}{n}}} \\
&= \frac{1}{\limsup |a_n|^{\frac{1}{n}}} \\
&= R
\end{aligned}$$

□

Example 3.4

$f(z) = \sum_{n=0}^{\infty} \frac{1}{n!} z^n$ then f is entire, $f'(z) = f(z)$ and $f(0) = 1$.

Corollary 3.7 **Smoothness of Power Series**

If f is represented by a power series on its disk of convergence, then derivatives of all orders exist on that disk.

3.1 Complex Integration

Definition 3.5 **Complex Integral**

Suppose $\varphi : [a, b] \to \mathbb{C}$ is piecewise continuous (continuous except at finitely many points where one-sided limits exist). Then $\operatorname{Re} \varphi$ and $\operatorname{Im} \varphi$ are also piecewise continuous. Set

$$\int_a^b \varphi := \int_a^b \operatorname{Re} \varphi + i \int_a^b \operatorname{Im} \varphi$$

Note this is complex linear in φ

Theorem 3.8 **Fundamental Theorem of Calculus**

If φ is piecewise differentiable (i.e. φ' is piecewise continuous) then

$$\int_a^b \varphi'(t) \, \mathrm{d}t = \varphi(b) - \varphi(a)$$

Theorem 3.9 **Integral Triangle Inequality**

$$\left|\int_a^b \varphi(t)\,\mathrm{d}t\right| \le \int_a^b |\varphi(t)|\,\mathrm{d}t$$